

Classical Features, Fine-Tuned Transformers, and Large Language Models for 20-Class Topic Classification on the 20 Newsgroups Corpus

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Abstract—This work presents a controlled empirical comparison of text-classification methods on the 20 Newsgroups corpus, a balanced 20-way topic-classification benchmark. Four methods are studied under a single preprocessing and evaluation pipeline. First, a fixed single-hidden-layer neural network is trained on four document representations: averaged word2vec embeddings trained for one and for thirty epochs, a TF-IDF vector, and an inverse-document- frequency-weighted word2vec vector. Second, `bert-base-uncased` is fully fine-tuned for classification. Third, a domain-adaptive masked-language- modelling (MLM) stage is run on the in-domain corpus before fine-tuning the same classifier. Fourth, the zero-shot and few-shot accuracy of a Llama-3 instruction model is compared against the fine-tuned BERT model on a common balanced 100-document subset. A bonus study adds a QLoRA parameter-efficient fine-tune of a 4-bit Llama-3.2-3B model on the same subset. On the full 7,532-document test set, the TF-IDF representation is the strongest classical baseline at 0.679 accuracy, while fine-tuned BERT reaches 0.690 and the MLM-adapted variant 0.691. On the 100-document subset, fine-tuned BERT (0.710) clearly outperforms Llama-3.1-8B zero-shot (0.620) and few-shot (0.440); QLoRA raises a 4-bit 3B model from 0.170 zero-shot to 0.530. The results indicate that, for this closed-set task, task-specific fine-tuning outperforms in-context learning, that representation quality dominates classical pipelines, and that domain-adaptive MLM yields only marginal gains on a small in-domain corpus.

Index Terms—Text classification, 20 Newsgroups, word embeddings, word2vec, TF-IDF, BERT, masked language modelling, domain adaptation, large language models, Llama-3, in-context learning, parameter-efficient fine-tuning, LoRA, QLoRA.

I. INTRODUCTION

TEXT classification is a core natural-language-processing task, and the 20 Newsgroups corpus is a long-standing benchmark for it [1]. The space of methods that can be applied to it now ranges from sparse lexical features and shallow neural networks, through pre-trained Transformer encoders, to billion-parameter instruction-tuned language models used without any task-specific training. These approaches differ enormously

in cost, data requirements, and inductive bias, and it is useful to see how they behave on exactly the same problem under exactly the same pipeline.

This report makes the following contributions. (i) It compares four document representations (word2vec at two training budgets, TF-IDF, and an IDF-weighted word2vec) using a single fixed classifier, isolating the effect of the representation. (ii) It fully fine-tunes `bert-base-uncased` for the same 20-way task. (iii) It evaluates whether a domain-adaptive masked-language-modelling stage on the in-domain corpus improves the fine-tuned classifier. (iv) It compares zero-shot and few-shot classification with a Llama-3 instruction model against fine-tuned BERT on a common balanced subset. (v) As a bonus study, it applies QLoRA parameter-efficient fine-tuning to a quantized Llama-3.2-3B model on the same subset. All experiments use a fixed random seed of 42 and were logged to Weights & Biases.

II. TOOLS AND REPRODUCIBILITY

All experiments were run in Python on a single NVIDIA L4 GPU (approximately 23 GB of memory) in a Google Colab environment. The dataset, the train/test split, the TF-IDF vectoriser, the PCA projection, and all metrics were provided by scikit-learn [2]. Word embeddings were trained with gensim [5] and text was tokenised with NLTK [6]. Neural training used PyTorch [11] with automatic mixed precision, and all Transformer and language models were loaded through the Hugging Face Transformers library [10]. The bonus study additionally used the PEFT and bitsandbytes libraries for low-rank adaptation and 4-bit quantization. Numerical reproducibility was sought by seeding Python, NumPy, and PyTorch with the same value of 42 throughout. Training and evaluation runs were tracked in the Weights & Biases projects `nalapro-20news` and `nalapro-bonus`; the public dashboards are available at <https://wandb.ai/raghav9raghav-hochschule-luzern/nalapro-20news> (Tasks 1–4) and <https://wandb.ai/raghav9raghav-hochschule-luzern/nalapro-bonus> (bonus task). The use of generative AI in preparing this report is described in the Generative AI Declaration.

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III. BACKGROUND AND RELATED WORK

Distributional word embeddings such as word2vec learn dense vectors in which geometric proximity reflects distributional similarity [4]; the skip-gram model and its gensim implementation are used here [5]. The TF-IDF weighting scheme remains a strong sparse baseline for topic classification [3]. The Transformer architecture [7] underlies BERT, a bidirectional encoder pre-trained with masked language modelling that is fine-tuned for downstream tasks [8]. Continuing the masked-language-modelling objective on in-domain text before fine-tuning (domain-adaptive pre-training) has been shown to help in many settings [9]. More recently, large autoregressive language models perform classification through zero-shot and few-shot in-context learning [13]; the Llama-3 family of instruction models is used here for that purpose [14]. Finally, parameter-efficient fine-tuning via low-rank adaptation (LoRA) [15] and its quantized variant QLoRA [16] makes it feasible to adapt such models on a single GPU.

IV. DATASET, PREPROCESSING, AND EXPLORATORY ANALYSIS

The 20 Newsgroups corpus was obtained through scikit-learn’s `fetch_20newsgroups` loader [1], [2] with the *headers*, *footers*, and *quotes* removed so that classification must rely on the message body rather than on metadata artefacts. The official split contains 11,314 training documents and 7,532 test documents across 20 classes. A stratified 90/10 split of the training portion produced 10,182 training and 1,132 validation documents; the official test set (7,532 documents) was held out unchanged.

The class distribution of the training and test portions is shown in Fig. 1. The corpus is approximately balanced, with most classes holding close to 600 documents in the training portion; the smallest classes are `talk.religion.misc`, `alt.atheism`, and `talk.politics.misc`. For the BERT experiments a per-class training cap of 600 documents was specified; because no class exceeded this cap after the validation split, the cap left the training set unchanged at 10,182 documents. The masked-language-modelling corpus used in Task 3 was the full 11,314-document training set. A small balanced evaluation set of 100 documents (5 per class) was constructed for the language-model experiments so that every method is scored on identical inputs.

For the Task 1 feature pipeline, documents were lower-cased, stripped of URLs, e-mail addresses, numerals, and non-word characters (apostrophes retained), tokenised with NLTK [6], and filtered to remove English stopwords and tokens shorter than three characters.

V. TASK 1: SHALLOW NEURAL NETWORK ON CLASSICAL REPRESENTATIONS

A. Method

A single fixed classifier was trained on four document representations to isolate the effect of the representation.

TABLE I

TASK 1 — SIMPLENN TEST RESULTS ON THE FULL 20-CLASS TEST SET (7,532 DOCUMENTS). BEST CLASSICAL RESULT IN BOLD.

Representation	Accuracy	Macro F1
word2vec (1 epoch)	0.272	0.236
word2vec (30 epochs)	0.642	0.625
TF-IDF	0.679	0.670
IDF-weighted word2vec (30 ep)	0.634	0.619

The classifier (*SimpleNN*) is a feed-forward network with one hidden layer of 128 units and a ReLU non-linearity, optimised with Adam [12] at a learning rate of 10^{-3} under a cross-entropy loss for 15 epochs, with a batch size of 64 (increased to 128 for the high-dimensional TF-IDF input).

The four representations are: (a) the mean of word2vec word vectors trained for a single epoch; (b) the mean of word2vec vectors trained for 30 epochs; (c) a TF-IDF vector; and (d) an IDF-weighted average of the 30-epoch word2vec vectors. Word2vec was trained with gensim [5] using the skip-gram objective, 100-dimensional vectors, a context window of 5, and a minimum count of 5, yielding a vocabulary of 17,296 words. The TF-IDF representation used unigrams, sublinear term-frequency scaling, a minimum document frequency of 2, a maximum document frequency of 0.9, and a vocabulary capped at 10,000 features, producing a $10,182 \times 10,000$ training matrix.

B. Results

Test-set results for all four representations are reported in Table I and summarised in Fig. 3. The single-epoch word2vec representation is essentially unusable, reaching only 0.272 accuracy, whereas 30 epochs of training raise accuracy to 0.642. The sparse TF-IDF representation is the strongest of the four at 0.679 accuracy and 0.670 macro F1. Weighting the 30-epoch word2vec vectors by IDF (0.634) did not improve on the simple mean (0.642). The effect of additional embedding training is also visible geometrically in Fig. 2: after a single epoch the topical probe words are largely intermixed, whereas after 30 epochs they form coherent topical clusters (vehicles, religion, computing, politics, medicine, space). It is worth noting that validation accuracy was consistently higher than test accuracy (for example, TF-IDF reached 0.768 on validation against 0.679 on test); this gap reflects the harder, metadata-stripped official test split rather than overfitting alone.

VI. TASK 2: FINE-TUNING BERT

A. Method

The `bert-base-uncased` encoder [8] was fully fine-tuned for sequence classification (109.5 M trainable parameters) using the Hugging Face Transformers library [10] on PyTorch [11]. Inputs were tokenised to a maximum length of 128. Training used the AdamW optimiser at a learning rate of 2×10^{-5} with a weight decay of 0.01, a linear schedule with a 10% warm-up ratio, gradient

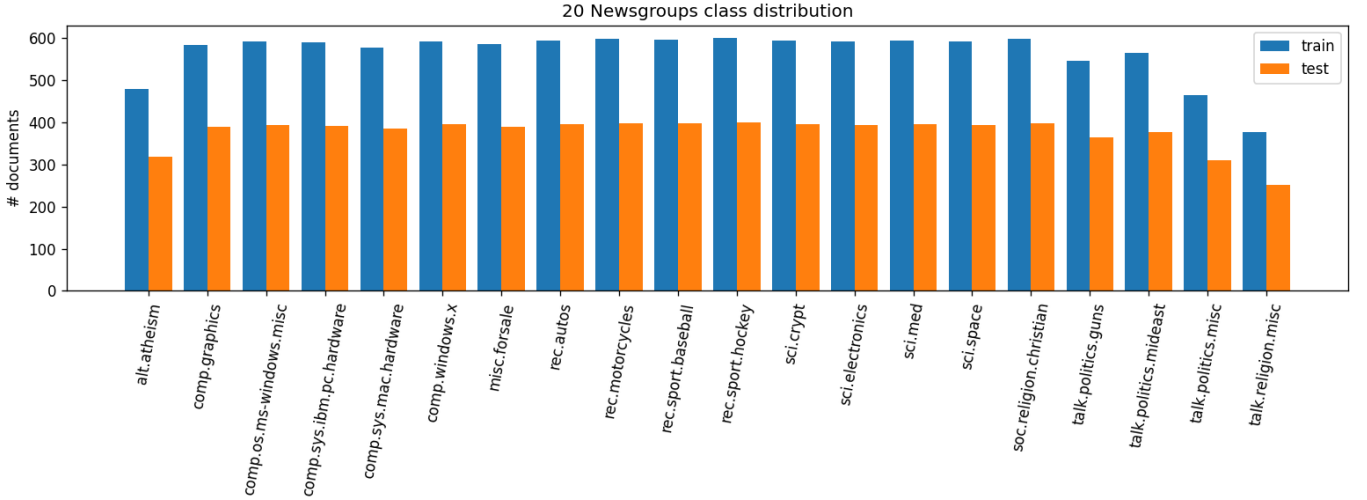


Fig. 1. Class distribution of the 20 Newsgroups training and test portions. The corpus is approximately balanced, with talk.religion.misc, alt.atheism, and talk.politics.misc the least represented.

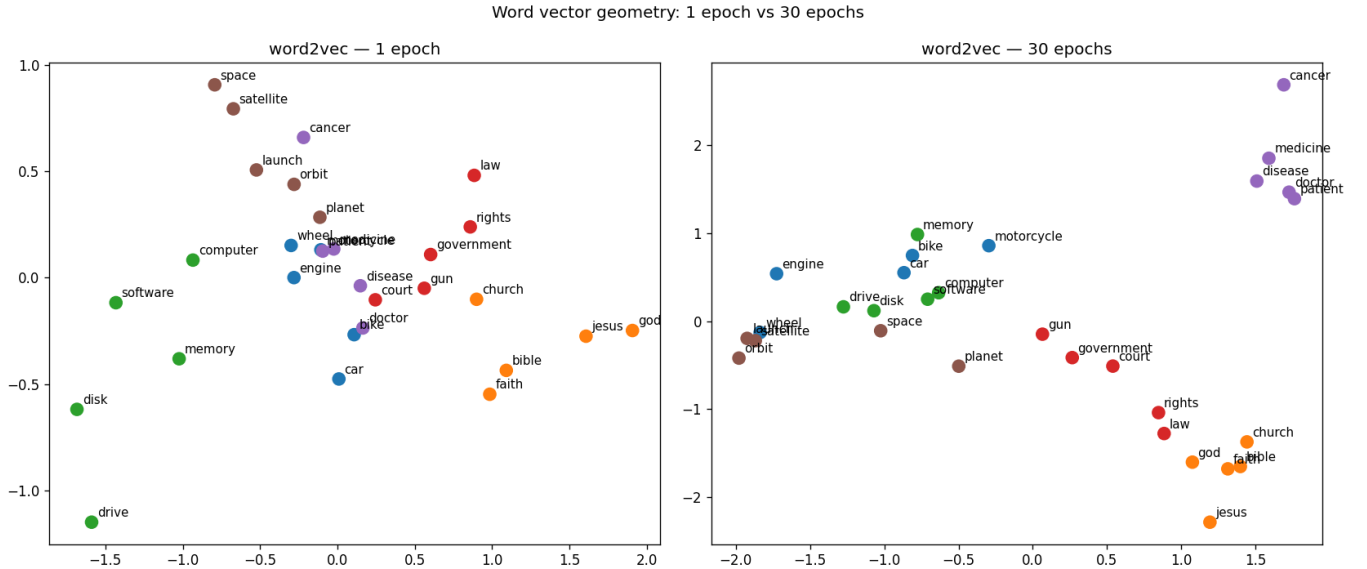


Fig. 2. Two-dimensional PCA projection of word2vec vectors for topical probe words after 1 epoch (left) and 30 epochs (right). Colours denote intended topic groups. Topical structure emerges only with sufficient training.

clipping at a norm of 1.0, a batch size of 32, and automatic mixed precision (fp16), for 3 epochs (957 optimiser steps; approximately 40 s per epoch on the L4 GPU).

B. Results

Validation accuracy rose monotonically across the three epochs (0.667 $\rightarrow$ 0.725 $\rightarrow$ 0.746). On the held-out test set the fine-tuned model reached an accuracy of 0.690 and a macro F1 of 0.665. The per-class behaviour is summarised in Table II and visualised in the confusion matrix of Fig. 4. The model is strongest on the sports, misc.forsale, and sci.med categories, and weakest on talk.religion.misc, which collapses to a macro F1 of 0.052 (recall 0.028) because its documents are overwhelmingly predicted as soc.religion.christian or

alt.atheism. The rec.autos class shows high recall but low precision, absorbing many neighbouring posts. These confusions are concentrated among the topically adjacent religion, politics, and computer-hardware groups.

VII. TASK 3: DOMAIN-ADAPTIVE MASKED LANGUAGE MODELLING

A. Method

This task tests whether adapting BERT to the corpus before classification helps. In the first stage the masked-language-modelling objective [8], [9] was continued on the full 11,314-document training corpus, using a masking probability of 0.15, a maximum length of 128, a batch size of 32, a learning rate of 5×10^{-5} , and a single epoch

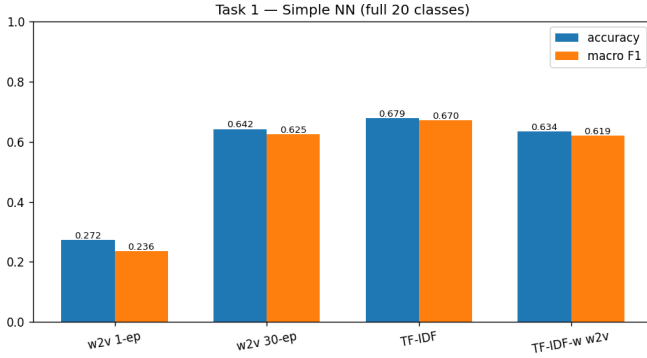


Fig. 3. Task 1 test accuracy and macro F1 for the four document representations. TF-IDF is the strongest classical baseline.

TABLE II

TASK 2 — PER-CLASS PRECISION, RECALL, AND F1 FOR FINE-TUNED BERT (THREE BEST AND THREE WORST CLASSES BY F1). MACRO AVERAGES AND ACCURACY ARE OVER ALL 20 CLASSES ON THE 7,532-DOCUMENT TEST SET.

Class	Precision	Recall	F1
rec.sport.hockey	0.877	0.860	0.868
rec.sport.baseball	0.893	0.821	0.856
sci.med	0.824	0.841	0.833
talk.politics.misc	0.470	0.503	0.486
alt.atheism	0.442	0.442	0.442
talk.religion.misc	0.350	0.028	0.052
Macro average	0.677	0.673	0.665
Overall accuracy			0.690

(354 steps); the mean MLM loss converged to 2.459. In the second stage the adapted weights were loaded into a sequence-classification head and fine-tuned with the same recipe as Task 2 (three epochs, learning rate 2×10^{-5}).

B. Results

Validation accuracy progressed as $0.695 \rightarrow 0.728 \rightarrow 0.737$. On the test set the MLM-adapted classifier reached an accuracy of 0.691 and a macro F1 of 0.668. As Fig. 5 shows, this is only marginally above the plain fine-tuned BERT of Task 2 (an improvement of about 0.1 accuracy point and 0.3 F1 point). The small effect is consistent with the corpus being modest in size and already in-domain for the eventual task, and with only a single MLM epoch being used; under these conditions domain-adaptive pre-training offers little headroom over direct fine-tuning.

VIII. TASK 4: LLAMA-3 IN-CONTEXT CLASSIFICATION VERSUS FINE-TUNED BERT

A. Method

This task compares in-context learning against task-specific fine-tuning on a common balanced 100-document subset (5 documents per class). The gated meta-llama/llama-3.2-3B-Instruct checkpoint was not accessible (HTTP 401), so the experiment fell back to the ungated NousResearch/Meta-Llama-3.1-8B-Instruct model

loaded in bfloat16 [14]. For zero-shot classification a system message enumerated the 20 permissible labels and instructed the model to emit exactly one of them; the post was truncated to 200 words, and the model decoded greedily for at most 20 new tokens, with the output mapped to the nearest label. For few-shot classification [13], one demonstration per class (20 demonstrations, truncated to 60 words each) was prepended. The fine-tuned BERT model from Task 2 was evaluated on the identical 100-document subset to provide a like-for-like comparison.

B. Results

Results are shown in Fig. 6. The fine-tuned BERT model is the strongest on this subset at 0.710 accuracy and 0.690 macro F1. Llama-3.1-8B zero-shot reached 0.620 accuracy and 0.637 macro F1, and few-shot *lowered* performance to 0.440 accuracy and 0.428 macro F1. On this closed-set 20-way task, task-specific fine-tuning therefore outperformed in-context classification, and adding a single generic demonstration per class hurt rather than helped, most plausibly because the long demonstration block introduced formatting noise relative to the clean zero-shot instruction.

IX. BONUS TASK: QLoRA FINE-TUNING OF A QUANTIZED LLAMA-3.2-3B

A. Method

The bonus study asks whether parameter-efficient fine-tuning can close the gap between an in-context language model and a fine-tuned encoder, at a fraction of the cost. The unsloth/llama-3.2-3B-Instruct model was loaded in 4-bit NF4 precision with double quantization and an fp16 compute type [14], [16]. Low-rank adapters [15] of rank 8 (scaling factor 16, dropout 0.05) were attached to the query, key, value, and output projection matrices, leaving 4,587,520 trainable parameters (approximately 0.14% of the model). Training used up to 100 documents per class (2,000 in total) for a single epoch with an effective batch size of 8 (batch size 2 with gradient accumulation of 4), a learning rate of 2×10^{-4} , the AdamW optimiser, and fp16 mixed precision; total training time was 9.27 minutes. Zero-shot, few-shot, and the QLoRA-adapted model were all evaluated on the same 100-document subset, with label parsing by exact substring match.

B. Results

The results are reported in Table III and Fig. 8. In this 4-bit 3B configuration, zero-shot classification reached only 0.170 accuracy (0.165 macro F1) and few-shot collapsed to 0.000, whereas QLoRA fine-tuning raised accuracy to 0.530 (0.502 macro F1) — more than tripling the zero-shot result for under ten minutes of training and roughly one-tenth of one percent of the parameters updated. The much lower zero-shot number than in Task 4 is expected: this

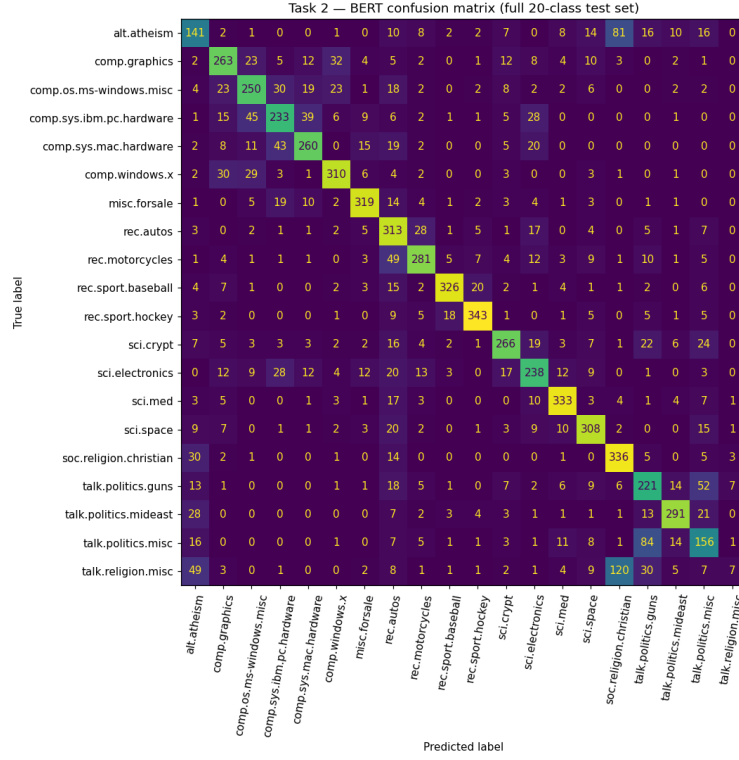


Fig. 4. Task 2 — confusion matrix of fine-tuned BERT on the full 20-class test set. Off-diagonal mass concentrates among topically adjacent religion, politics, and computer-hardware categories.

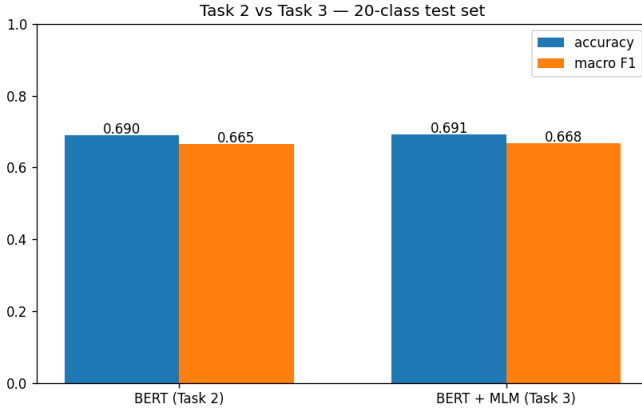


Fig. 5. Task 2 versus Task 3 on the full 20-class test set. The domain-adaptive MLM stage yields only a marginal improvement over direct fine-tuning.

experiment uses a smaller, more aggressively quantized model (3B at 4-bit versus 8B at bfloat16), a plain-text prompt rather than a chat template, and an exact-substring label parser that returns “unknown” when the model does not copy a label verbatim. The few-shot collapse to zero accuracy follows directly from that parser: under the long few-shot prompt the model produced no exactly matching label string on any evaluation document. These differences mean the Task 4 and bonus in-context numbers should not be read as directly comparable.

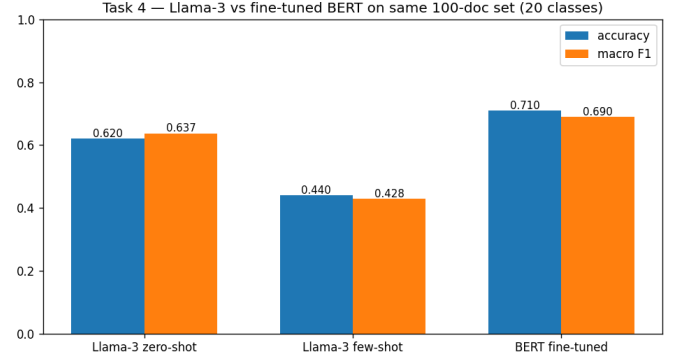


Fig. 6. Task 4 — Llama-3.1-8B zero-shot and few-shot versus the fine-tuned BERT model from Task 2, all evaluated on the same 100-document subset (20 classes). Fine-tuned BERT is strongest; few-shot underperforms zero-shot.

TABLE III
BONUS TASK — ZERO-SHOT, FEW-SHOT, AND QLoRA FINE-TUNING OF A 4-BIT LLAMA-3.2-3B MODEL ON THE SAME 100-DOCUMENT SUBSET. EVALUATION RUNTIME IN SECONDS.

Method	Accuracy	Macro F1	Runtime (s)
Zero-shot	0.170	0.165	80
Few-shot (1/class)	0.000	0.000	102
QLoRA fine-tuned	0.530	0.502	111

X. OVERALL COMPARISON AND DISCUSSION

Table IV consolidates every method, and Fig. 7 plots their accuracies side by side. Five lessons emerge. First, for

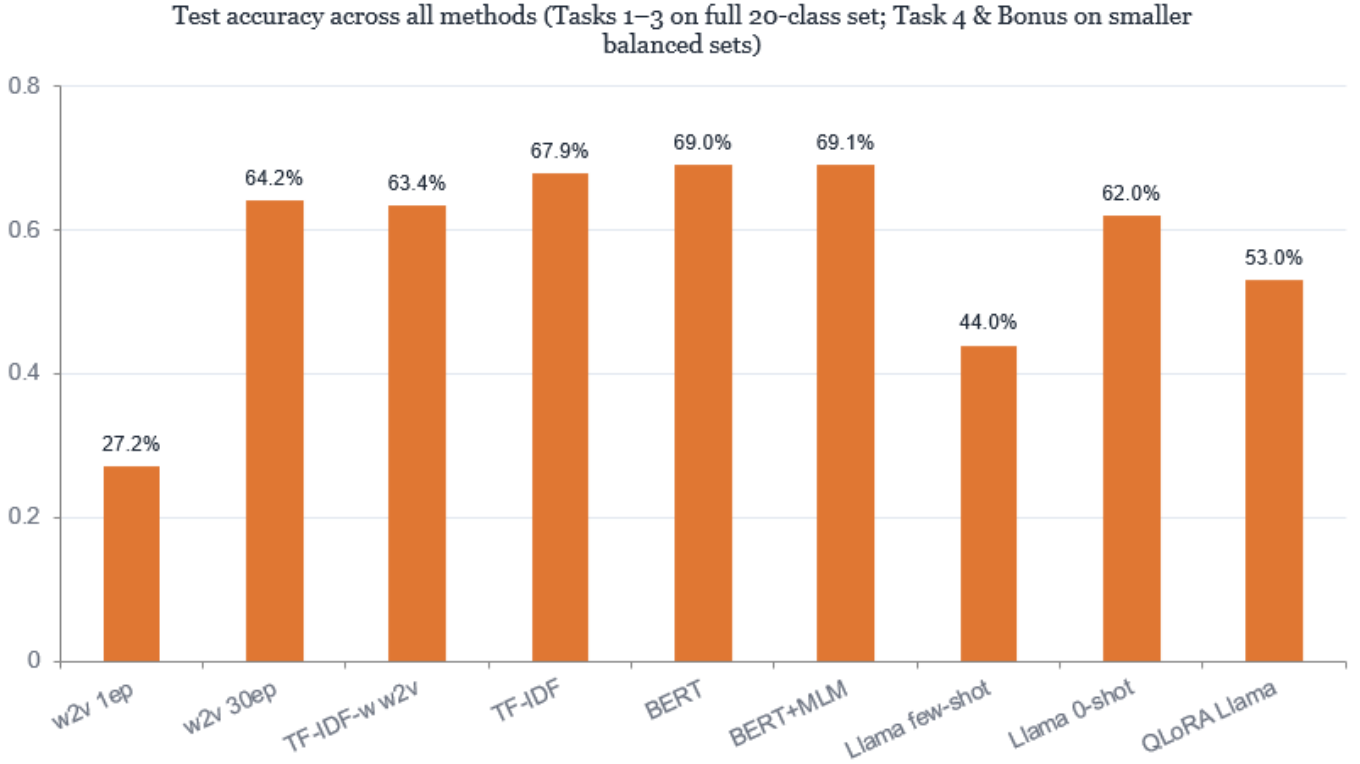


Fig. 7. Test accuracy across all methods. Tasks 1–3 (the four SimpleNN feature variants, fine-tuned BERT, and BERT with domain-adaptive MLM) are scored on the full 7,532-document test set; the Task 4 Llama-3 runs (zero-shot and few-shot) and the bonus QLoRA run are scored on smaller balanced subsets, so accuracies are not directly comparable across the two groups.

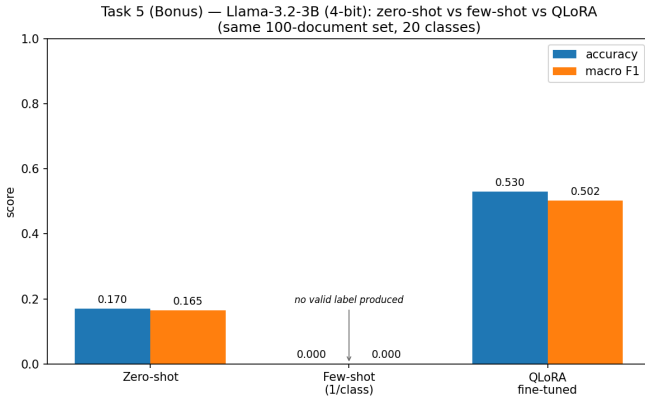


Fig. 8. Task 5 (bonus) — accuracy and macro F1 of the 4-bit Llama-3.2-3B model under zero-shot, few-shot, and QLoRA fine-tuning on the same 100-document subset. Few-shot collapses to zero because the model produced no exactly matching label string on any evaluation document; QLoRA more than triples the zero-shot accuracy.

the classical pipeline the representation matters far more than the classifier: with a fixed network, an undertrained embedding is almost useless (0.272), thirty epochs of word2vec recover most of the performance (0.642), and a simple TF-IDF vector is the strongest of all (0.679). Second, fine-tuned BERT is the best method overall, leading on the full test set (0.690) and on the 100-document subset (0.710). Third, domain-adaptive masked language modelling gives only a marginal gain here (0.691 versus

TABLE IV

CONSOLIDATED RESULTS ACROSS ALL METHODS. ROWS ON THE FULL TEST SET USE 7,532 DOCUMENTS; ROWS ON THE SUBSET USE THE SAME BALANCED 100 DOCUMENTS. ALL VALUES ARE TAKEN DIRECTLY FROM THE PROJECT’S LOGGED OUTPUTS.

Method	Model / representation	Acc.	Macro F1
<i>Full 20-class test set (7,532 docs)</i>			
SimpleNN	word2vec (1 epoch)	0.272	0.236
SimpleNN	word2vec (30 epochs)	0.642	0.625
SimpleNN	TF-IDF	0.679	0.670
SimpleNN	IDF-weighted word2vec	0.634	0.619
Fine-tune	BERT (bert-base)	0.690	0.665
Fine-tune	BERT + domain MLM	0.691	0.668
<i>Balanced subset (100 docs)</i>			
Zero-shot	Llama-3.1-8B	0.620	0.637
Few-shot	Llama-3.1-8B (1/class)	0.440	0.428
Fine-tune	BERT (on same subset)	0.710	0.690
Zero-shot	Llama-3.2-3B (4-bit)	0.170	0.165
Few-shot	Llama-3.2-3B (4-bit)	0.000	0.000
QLoRA	Llama-3.2-3B (4-bit)	0.530	0.502

0.690), so it was not worth its extra stage on this small in-domain corpus. Fourth, for this closed-set 20-way task, task-specific fine-tuning beats zero-shot and few-shot in-context learning, and naive few-shot prompting can actively hurt. Fifth, parameter-efficient fine-tuning is an attractive middle ground: QLoRA more than tripled the zero-shot accuracy of a quantized 3B model while updating only about 0.14% of its parameters in roughly nine minutes. A clear caveat is that the language-model

comparisons are run on only 100 documents and are therefore high-variance, and that prompt formatting and label-parsing choices visibly shift the reported in-context scores.

XI. LIMITATIONS AND FUTURE WORK

Several limitations should temper these conclusions. The language-model evaluation set is only 100 documents, so its accuracy estimates carry large confidence intervals. Every experiment was run with a single random seed, without reporting variance. The domain-adaptive stage used a single MLM epoch, and hyperparameter search throughout was minimal. The Task 4 and bonus in-context results were obtained with different Llama variants (8B at bfloat16 versus 3B at 4-bit), different prompt templates, and different label parsers, which makes them not directly comparable to each other. Few-shot prompting used a single generic demonstration per class, and no structured-decoding or constrained-generation method was applied to force valid labels. Finally, beyond wall-clock runtimes, no systematic study of inference cost or latency was conducted. Future work could enlarge the language-model evaluation set, report results over multiple seeds with confidence intervals, train the MLM stage for longer, increase the maximum sequence length, use stronger and constrained prompts, and hold the model family fixed across the in-context and fine-tuning comparisons.

XII. CONCLUSION

Across a spectrum of methods applied to the same 20-class topic-classification problem, fine-tuned BERT was the most accurate, both on the full test set and on a common subset. A simple TF-IDF representation proved to be a strong and cheap classical baseline, while undertrained word embeddings were ineffective. Domain-adaptive masked language modelling produced only a marginal gain on this small in-domain corpus. In-context classification with a Llama-3 instruction model underperformed task-specific fine-tuning and was sensitive to prompt and parsing design, but QLoRA parameter-efficient fine-tuning recovered much of the gap at a small fraction of the cost. For closed-set topic classification of this kind, the evidence here favours fine-tuning a compact encoder, with parameter-efficient adaptation of a generative model as a practical alternative.

CODE AND DATA AVAILABILITY

The code for all tasks is available at <https://github.com/ss0109/nalapro-project>, and the training and evaluation runs are logged to the Weights & Biases projects `nalapro-20news` and `nalapro-bonus` at <https://wandb.ai/raghav9raghav-hochschule-luzern/nalapro-20news> and <https://wandb.ai/raghav9raghav-hochschule-luzern/nalapro-bonus> respectively. The 20 Newsgroups dataset is not committed to the repository; it is downloaded at run time through scikit-learn’s `fetch_20newsgroups` loader [1], [2].

GENERATIVE AI DECLARATION

Generative AI tools were used as a writing and coding aid in the preparation of this report and the accompanying code, for example to draft and refine prose, to help structure the implementation, and to debug. The experimental design, the choice of methods and hyperparameters, the execution of all training and evaluation, the numerical results, and their interpretation are the author’s own work. All reported figures and metrics were produced by the author’s own experiments, and the author takes full responsibility for the content of this report.

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